

Crossdating events, ecological meaning and recognition evidence

Lag signatures are read from older years toward the bark-side end.

Event and executable edit	Lag signature	Ecological / specimen meaning	Recognition evidence
Clean match Keep the series unchanged	Stable 0 (or a known global baseline)	Calendar alignment is retained across the measured series.	No persistent lag transition; edit counterfactuals provide no stable gain; reference cores agree on the baseline.
Missing ring Insert one calendar year	$g - 1 \rightarrow g$	A locally absent or extremely narrow ring, often associated with suppressed cambial growth under drought, cold, fire or insect stress.	A unit negative step, a superior insert-year counterfactual and concordant gains against the master and independent cores.
False ring Delete one calendar year	$g + 1 \rightarrow g$	An intra-annual density fluctuation or false boundary, including a ring divided twice during measurement.	A unit positive step, a superior delete-year counterfactual, residual support and consistent transition direction across references.
Partial shift Move the older block by $-q$ years	$g - q \rightarrow g$ $q \geq 2$	Localized decay, fracture, segment loss or a within-core measurement splice that displaces only part of the series.	One large local transition, a dominant range-shift counterfactual and agreement on the fixed-side breakpoint.
Whole-series shift Move the complete series by g years	Constant global g	A negative shift is consistent with tree death, missing bark or sapwood, or a bark-side break that removed outer years.	Global sliding match, a constant lag path, terminal-side evidence and a whole-series counterfactual converge on the same g .
Event staircase Apply sequential unit edits	$-n \dots -1 \rightarrow 0$ (or the positive mirror)	Several missing or false rings distributed through the series rather than one continuous displaced block.	Multiple intermediate lag plateaus, repeated unit counterfactual gains and stepwise recovery after each confirmed edit.
Adjacent paired events Resolve two local unit events	Short pulse; distal baseline restored	Nearby missing and false rings can cancel in long windows while retaining a distinct local disturbance signature.	Opposite local transitions, paired before/after evidence and matching votes on both sides of the short event interval.